

## ASSIGNMENT – 2(technical presentation)

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**Course code: csa1613**

**Course name: datawarehouse and datamining**

## Topic 2: OLTP vs OLAP

## Online transaction processing (OLTP):

- Handles daily transactions
- Fast processing
- Current data
- Example:
  - ATM
  - Online shopping

## Online analytical processing (OLAP):

- Supports analysis
- Historical data
- Decision making
- Forecasting



**OLTP vs OLAP**

|                        |                 |
|------------------------|-----------------|
|                        |                 |
| <b>OLTP</b>            | <b>OLAP</b>     |
| Transaction processing | Data analysis   |
| Current data           | Historical data |
| Many users             | Few analysts    |
| Fast updates           | Complex queries |

**Key Differences**

- Purpose
- Users
- Data
- Speed
- Query complexity

**Performance Comparison:**

OLTP

- Thousands of transactions

OLAP

- Large analytical queries
- Multi-dimensional analysis

## **Use Cases:**

### OLTP

- Banking
- Hospital billing

### OLAP

- Sales analysis
- Business forecasting

## **Integration:**

- OLTP collects data
- ETL transfers data
- Data warehouse stores data
- OLAP analyzes data

## **Example:**

### Amazon

- OLTP → Customer Orders
- OLAP → Sales Analysis

## **Slide 10: Conclusion**

Both systems are important and complement each other.